

Event Management Database - SQL Exercise Solutions

1. User Upcoming Events

```
SELECT u.full_name, e.title, e.city, e.start_date
FROM Users u
JOIN Registrations r ON u.user_id = r.user_id
JOIN Events e ON r.event_id = e.event_id
WHERE e.status='upcoming' AND u.city=e.city
ORDER BY e.start_date;
```

2. Top Rated Events

```
SELECT e.event_id, e.title, AVG(f.rating) AS avg_rating
FROM Events e
JOIN Feedback f ON e.event_id=f.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(*) >= 10;
```

3. Inactive Users

```
SELECT *
FROM Users
WHERE user_id NOT IN (
    SELECT DISTINCT user_id
    FROM Registrations
    WHERE registration_date >= CURDATE() - INTERVAL 90 DAY
);
```

4. Peak Session Hours

```
SELECT e.title, COUNT(*) AS session_count
FROM Events e
JOIN Sessions s ON e.event_id=s.event_id
WHERE TIME(s.start_time) BETWEEN '10:00:00' AND '12:00:00'
GROUP BY e.event_id, e.title;
```

5. Most Active Cities

```
SELECT u.city, COUNT(*) AS registrations
FROM Users u
JOIN Registrations r ON u.user_id=r.user_id
GROUP BY u.city
ORDER BY registrations DESC
LIMIT 5;
```

6. Event Resource Summary

```
SELECT e.title, COUNT(r.resource_id) AS resources
FROM Events e
LEFT JOIN Resources r ON e.event_id=r.event_id
GROUP BY e.event_id, e.title;
```

7. Low Feedback Alerts

```
SELECT u.full_name, e.title, f.comments
FROM Feedback f
JOIN Users u ON f.user_id=u.user_id
JOIN Events e ON f.event_id=e.event_id
WHERE f.rating < 3;
```

8. Sessions per Upcoming Event

```
SELECT e.title, COUNT(s.session_id) AS session_count
FROM Events e
LEFT JOIN Sessions s ON e.event_id=s.event_id
WHERE e.status='upcoming'
GROUP BY e.event_id, e.title;
```

9. Organizer Event Summary

```
SELECT u.full_name, e.status, COUNT(*) AS total_events
FROM Users u
JOIN Events e ON u.user_id=e.organizer_id
GROUP BY u.full_name, e.status;
```

10. Feedback Gap

```
SELECT e.title
FROM Events e
JOIN Registrations r ON e.event_id=r.event_id
LEFT JOIN Feedback f ON e.event_id=f.event_id
GROUP BY e.event_id, e.title
HAVING COUNT(f.feedback_id)=0;
```

11. Daily New User Count

```
SELECT registration_date, COUNT(*) FROM Users GROUP BY registration_date;
```

12. Event with Maximum Sessions

```
SELECT event_id, COUNT(*) FROM Sessions GROUP BY event_id ORDER BY COUNT(*) DESC;
```

13. Average Rating per City

```
SELECT e.city, AVG(f.rating) FROM Events e JOIN Feedback f ON e.event_id=f.event_id
GROUP BY e.city;
```

14. Most Registered Events

```
SELECT event_id, COUNT(*) FROM Registrations GROUP BY event_id ORDER BY COUNT(*)  
DESC LIMIT 3;
```

15. Event Session Time Conflict

```
SELECT * FROM Sessions s1 JOIN Sessions s2 ON s1.event_id=s2.event_id AND  
s1.session_id<s2.session_id;
```

16. Unregistered Active Users

```
SELECT * FROM Users WHERE registration_date>=CURDATE()-INTERVAL 30 DAY;
```

17. Multi-Session Speakers

```
SELECT speaker_name, COUNT(*) FROM Sessions GROUP BY speaker_name HAVING  
COUNT(*)>1;
```

18. Resource Availability Check

```
SELECT e.title FROM Events e LEFT JOIN Resources r ON e.event_id=r.event_id WHERE  
r.resource_id IS NULL;
```

19. Completed Events with Feedback Summary

```
SELECT e.title, AVG(f.rating) FROM Events e JOIN Feedback f ON e.event_id=f.event_id  
WHERE e.status='completed' GROUP BY e.title;
```

20. User Engagement Index

```
SELECT u.full_name, COUNT(DISTINCT r.event_id), COUNT(DISTINCT f.feedback_id) FROM  
Users u LEFT JOIN Registrations r ON u.user_id=r.user_id LEFT JOIN Feedback f ON  
u.user_id=f.user_id GROUP BY u.user_id;
```

21. Top Feedback Providers

```
SELECT user_id, COUNT(*) FROM Feedback GROUP BY user_id ORDER BY COUNT(*) DESC  
LIMIT 5;
```

22. Duplicate Registrations Check

```
SELECT user_id,event_id,COUNT(*) FROM Registrations GROUP BY user_id,event_id  
HAVING COUNT(*)>1;
```

23. Registration Trends

```
SELECT YEAR(registration_date), MONTH(registration_date), COUNT(*) FROM  
Registrations GROUP BY YEAR(registration_date), MONTH(registration_date);
```

24. Average Session Duration per Event

```
SELECT event_id, AVG(TIMESTAMPDIFF(MINUTE,start_time,end_time)) FROM Sessions  
GROUP BY event_id;
```

25. Events Without Sessions

```
SELECT e.title FROM Events e LEFT JOIN Sessions s ON e.event_id=s.event_id WHERE  
s.session_id IS NULL;
```